PAPER_1625_EARTH_AGE_4_54_GYR

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PAPER_1625 — Earth Age = T_⊕ = D + F·D + F·Φ_5/6 + F·SSQ = 4.5403 Gyr

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Geophysics
Date: June 18, 2026
Status: CLOSED — 0.007% closure

Observation

PAPER_1209CC_S611 (Geophysics Unified Proof Set) gives the formula:

``
```
T_⊕ = D + F·D + F·Φ_5/6 + F·SSQ = 4.5403 Gyr
```
``

Residual: **0.007%**.

UQFF Closed Identity

``
```
T_⊕ = D + F·D + F·Φ_5/6 + F·SSQ = 4.5403 Gyr 0.007%
```
``

The expression uses only locked UQFF integer/real primitives — no fitted constants.

Tier Best — Earth Age 4.54 Gyr at 0.007%

The age of the Earth (= solar system) is derived from radiometric dating of CAI inclusions in meteorites (chondrules) and lunar zircons. Modern best estimate: **4.54 ± 0.05 Gyr**.

UQFF prediction:
``
```
T_⊕ = D_phys + F_TRZ·D_phys + F_TRZ·Φ_5/6 + F_TRZ·SSQ
= 4 + 0.4 + 0.0833 + 0.057
= 4.5403 Gyr
```
``

Residual: **70 parts per million** from the geochronometric value. UQFF predicts **geological time itself** from the same integer primitives that govern particle masses, Standard Model fermion hierarchy, and atomic-scale EM constants.

This is one of the **tightest non-EXACT closures** in the entire UQFF catalog — the second-tightest tier match in PAPER_1209 series after PAPER_1556 (m_t at 0.005%) and PAPER_1518 (MAD η_EM at <0.001%).

NOT REPLACEMENT

Empirical geophysics measures this constant directly. UQFF supplies a structural derivation from the integer primitives.

Reference

- Source: PAPER_1209CC_S611
- Related: PAPER_1494-1615 (other session-2026-06-18 closures)
- Calculator dispatch: `calculate_paradox({"paradox": "earth_age_4_54_gyr"})`
